

Joohyun Lee

✉ jin.joohyun.lee@gmail.com | ☎ +82-10-9286-9766 | 🌐 jin-joohyun-lee.github.io | 🆔 0009-0001-5935-4220

RESEARCH INTERESTS

Organic Electrochemical Devices (OECTs, OECs), Conjugated Polymer Synthesis, Bioelectronic Applications

EDUCATION

Chung-Ang University (CAU)

- M.S. in Chemical Engineering
GPA: 4.18 / 4.5
- B.S. in Chemical Engineering and Materials Science
GPA: 3.75 / 4.5
Leave of absence for mandatory military service (May 2019–Dec. 2020)

Seoul, Republic of Korea

Sep. 2024–Present

Advisor: Prof. Felix Sunjoo Kim

Mar. 2018–Aug. 2024

Advisor: Prof. Felix Sunjoo Kim

PUBLICATIONS

1. “Morphology and Crystallinity of Transparent Capacitive Ion Storage Layer for Electrochromic Displays.”
 - Seungwon Lee, Yuseok Song, **Joohyun Lee**, Jaeyong Ahn, Won-June Lee, Jiao Suo, Yoonhoo Lee, Liyan You, Jianguo Mei, Felix Sunjoo Kim, and Inho Song*.
 - *ACS Applied Materials & Interfaces*, accepted (2026).
2. “Enhancement-Mode PEDOT:PSS Electrochemical Transistors with Solid Electrolytes Based on [DBUH][OAc] as a Dual-Role Ionic Component.”
 - Donguk Kim[†], **Joohyun Lee**[†], Seungjin Song, Youngseok Kim, and Felix Sunjoo Kim*.
 - Under review.
3. “Electrolyte-Driven Threshold Voltage Modulation in PEDOT:PSS OECTs via Anion Control of DBU-based Acid-Base Ionic Liquids.”
 - **Joohyun Lee**[†], Donguk Kim[†], Youngseok Kim, Seungjin Song, and Felix Sunjoo Kim*.
 - Manuscript in preparation; submission expected in 2026.
4. “Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering and Oxidative Polymerization.”
 - **Joohyun Lee**, Seungjin Song, and Felix Sunjoo Kim*.
 - In preparation.

RESEARCH EXPERIENCE

Prof. Felix Sunjoo Kim Lab, Chung-Ang University

Graduate Researcher

Seoul, Republic of Korea

Sep. 2024–Present

- **Project 1:** Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering and Oxidative Polymerization
 - Synthesized EDOT-based D-A-D monomers via Stille coupling (confirmed by ¹H NMR) and exploited the high reactivity of the α -site of EDOT for facile oxidative polymerization via aqueous-phase (PSS-templated) and surface methods, adding a base to retard the rapid polymerization during spin-coating
 - Determined the energy levels of synthesized monomers and polymers by correlating optical bandgap (UV-Vis-NIR) with electrochemical measurements (CV) and DFT calculations (ORCA)
- **Project 2:** Electrolyte-Driven Threshold Voltage Modulation in PEDOT:PSS OECTs via Anion Control of DBU-based Acid-Base Ionic Liquids
 - Controlled the ionicity of DBU-based ABILs by systematically varying the pKa of carboxylic acid components used in synthesis, characterized via ¹H NMR spectroscopy and DFT calculations
 - Characterized threshold voltage shift in PEDOT:PSS OECTs as a function of ABIL ionicity-dependent de-doping level via chemical (UV-Vis-NIR, Raman, XPS), electrical (transfer/output curves), and electrochemical (CV, EIS) characterization
 - Implemented zero- V_{GS} -load OECT inverters by exploiting the ABIL-tuned threshold voltage

Prof. Felix Sunjoo Kim Lab, Chung-Ang University

Undergraduate Researcher

Seoul, Republic of Korea

Mar. 2022–Aug. 2024

- **Project 1:** Enhancement-Mode PEDOT:PSS Electrochemical Transistors with Solid Electrolytes Based on [DBUH][OAc] as a Dual-Role Ionic Component
 - Demonstrated that [DBUH][OAc] exhibits distinct interaction with PEDOT:PSS compared to [EMIM][TFSI], which is critical for enhancement-mode OECT operation
 - Revealed via DFT and XPS that [DBUH][OAc] preferentially interacts with the ion-conduction pathway in dry PEDOT:PSS, yielding C-V profiles with distinct accumulation and depletion regions analogous to organic semiconductors

- **Project 2: Long-Term Memory Organic Transistor Utilizing Electrolyte and Ferroelectric Polymer Bilayer Gate Dielectric**
 - Demonstrated synaptic memory with >24-hour retention and lower operating voltage in bilayer-gated organic transistors compared to ferroelectric polymer-only counterparts
 - Initially attributed the memory to ferroelectric polarization based on P-E hysteresis of the bilayer, but revealed through control experiments without the ferroelectric polymer that the long-term memory originated from the increased ionic resistance of the low-ion-concentration electrolyte (in collaboration with H. Yang)

FELLOWSHIPS, AWARDS & HONORS

- **BK21 FOUR Fellowship**, Ministry of Education, Korea (monthly stipend, ~\$800) Sep. 2024–Aug. 2026
- **Research Assistantship**, CAU (full tuition waiver, ~\$6,000/semester) Fall 2025
- **Teaching Assistantship**, CAU Sep. 2024–Aug. 2025
- **Academic Excellence Scholarship**, CAU (ranked 1st in department) Spring 2024
- **Undergraduate Research Scholarship**, CAU (Practical & Research Engagement program; selected twice) Sep. 2022–Feb. 2024
- **Leadership Scholarship**, CAU (American Football Club president & captain) Mar. 2021–Feb. 2023

SELECTED CONFERENCES

1. “Switching from Depletion- to Enhancement-mode in PEDOT:PSS-based Electrochemical Transistors.”
Donguk Kim, **Joohyun Lee**, Jihoon Kim, and Felix Sunjoo Kim*.
ITC 2024, 18th International Thin-Film Transistor Conference, 2024.
2. “Threshold Voltage Modulation of PEDOT:PSS Based Electrolyte-Gated Transistors by Anion Control in Acid-Base Type Ionic Liquid.”
Joohyun Lee, Donguk Kim, Youngseok Kim, and Felix Sunjoo Kim*.
FPI 16 Symposium, 16th International Symposium on Functional π -Electron Systems, 2025.
3. “Non-Volatile Switching Characteristics of Organic Electrochemical Transistors Enabled by Extremely Low Ion Concentration of Gel Electrolyte.”
Heewon Yang, Jungmin Lee, **Joohyun Lee**, Minuk Kim, Hyejoon Park, Donguk Kim, Youngseok Kim*, and Felix Sunjoo Kim*.
FPI 16 Symposium, 16th International Symposium on Functional π -Electron Systems, 2025.
4. “Toncity of Protic Ionic Liquid Influences Channel Conductance of Conducting Polymer in Electrochemical Transistors.”
Joohyun Lee, Donguk Kim, and Felix Sunjoo Kim*.
2026 KICHe Spring Meeting, 2026.
5. “Systematic Tuning of Energy Levels in EDOT-Based D-A-D Monomers via Acceptor Engineering.”
Joohyun Lee, Seungjin Song, and Felix Sunjoo Kim*.
KJF-ICOME 2026, 2026 KJF-International Conference on Organic Materials for Electronics and Photonics, 2026.

EXPERIENCE & ACTIVITIES

- **Teaching Assistant**, Department of Chemical Engineering, CAU Sep. 2024–Aug. 2025
Special Topics in Chemical Engineering; Chemical Process Laboratory (process simulation with Aspen HYSYS)
- **Internship**, MeasHub Jul. 2026–Sep. 2026
Electrical & electrochemical measurement platform where researchers share recipes and store results in the cloud. Supported program planning and coordination, and contributed to organizing and running conferences on site
- **President & Captain**, CAU American Football Club Mar. 2018–Feb. 2023
Team member from 2018; president and captain (2021–2023), leading club operations and on-field performance. Seoul Collegiate American Football Tournament: Champion (2018), Runner-up (2019), 3rd Place (2021)

SKILLS

- **Device & Electrochemical Characterization**: OECT fabrication and figure-of-merit extraction (g_m , V_T , on/off, μC^* , transient), OECT inverters (gain, noise margin), CV, EIS, C–V (C^*), spectroelectrochemistry [HP4156A, Ki2634B, SP-150e]
- **Synthesis**: Stille coupling, oxidative polymerization (solution & surface), protic ionic liquid synthesis, column chromatography
- **Structural & Spectroscopic**: XPS, Raman, UV-Vis-NIR, FT-IR, NMR
- **Morphological & Thermal**: AFM, TGA, DSC
- **Computational**: DFT/TD-DFT (ORCA, Spartan), Python, MATLAB
- **Languages**: Korean (Native), English (Proficient)

REFERENCES

Dr. Felix Sunjoo Kim, Professor, Chung-Ang University
 Dr. Inho Song, Assistant Professor, Chung-Ang University
 Dr. Youngseok Kim, MeasHub (internship supervisor)

fskim@cau.ac.kr
 ihsong@cau.ac.kr
 youngs.kim@outlook.com